Do Now · retrieval

Lesson 091 · Fetch–Decode–Execute: A Register Trace

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which register holds the current instruction?

Q2

What is the role of MAR?

Q3

Where is the next instruction address kept?

Q4

A full adder has A=1,B=0 and carry-in=1. State Sum and carry-out.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Fetch copies the address from PC to MAR; memory read transfers the instruction through MDR into CIR. In a sequential fetch, PC advances.

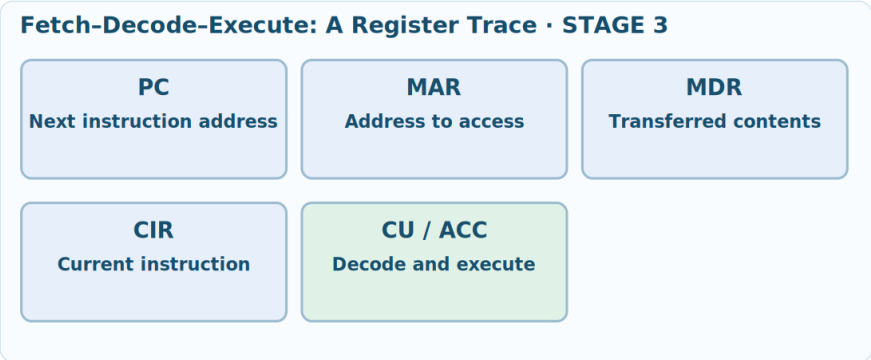
Core idea 2

Explain — The Control Unit decodes the current instruction held in CIR and produces the necessary control signals.

Core idea 3

Evaluate — Execution of LDA 80 reads the data held at memory address 80 into ACC; the next instruction is not executed until its own fetch.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

fetch — Retrieve next instruction.

decode — CU interprets instruction.

execute — Carry out operation.

CIR — Current Instruction Register.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

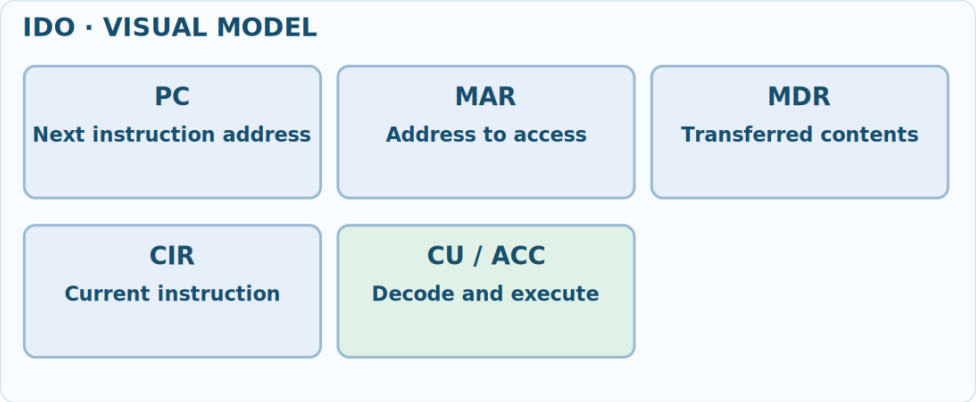
Problem

PC=30, memory[30]=ADD 70, memory[70]=5, ACC=4. Trace after fetch and execute.

Follow the worked method

1. Explain the mechanism — Fetch: MAR30, CIR ADD70, PC31. Execution reads memory[70]=5 and ALU adds to ACC4, giving ACC9. — The PC increments in fetch while the ALU operates on ACC and fetched operand.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the mechanism
After fetch PC43,CIR SUB90. Operand read from90 into MDR=3; ALU computes11−3, ACC8.
The addressed operand must not be confused with instruction address42.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace the fetch of instruction at PC20 into MAR,MDR,CIR; give PC after sequential increment. [6]

Question 2

Explain difference between decode and execute for LDA 80 when memory[80]=13. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

An answer states 'MDR stores address 80 and MAR stores the fetched 13'. Correct the error with a precise memory-read explanation. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

During a sequential fetch, MDR stores the instruction transferred from memory rather than its address.

Use key vocabulary from memory

1. Explain fetch in your own words.

2. Explain decode in your own words.

3. Explain execute in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.